

Jackson Bopp

941-456-5553 | boppjackson@gmail.com | linkedin.com/in/jbopp | github.com/JacksonBopp | Tampa, FL (Open to Relocate)

Education

University of South Florida

B.S. Computer Engineering, Dean's List, Gold's Scholars | GPA: 3.52

Tampa, FL

Fall 2026

Florida SouthWestern State College

High School Diploma, Associate in Arts, Dean's List | GPA: 3.86

Punta Gorda, FL

Fall 2022

Projects

Wireless Motor PWM Controller

Summer 2026

MSP430FR2355, Bare-Metal C, HM-10 BLE, L298N, I2C, UART, PWM, Timer ISRs

- Collaborated with a 4-member team to build bare-metal C firmware for the MSP430FR2355 that reads motor commands from a 4x4 matrix keypad and an HM-10 BLE module and drives a DC motor through an L298N H-bridge.
- Wrote an interrupt-driven UART receiver for the BLE module with a ring buffer, so wireless and physical keypad input run through the same command dispatcher and LCD status display.
- Generated an 8 kHz PWM signal on Timer B2 and added a kickstart pulse plus duty cycle remapping to compensate for the L298N's voltage drop and the motor's static friction.
- Used ACLK-based timers for the HD44780 LCD (via a PCF8574 I2C backpack) and keypad scan delays to keep timing accurate across DCO frequency drift.
- Documented the system with a full wiring diagram, system block diagram, and demo video as part of the final project report.

Testbench

Summer 2026

MSP430FR2355, ESP32, STM32F103C8, RP2040, MQTT, Raspberry Pi, Docker

- Built a hardware-agnostic test platform that streams live telemetry from UART-based microcontrollers through a Raspberry Pi bridge over MQTT to a web dashboard.
- Wrote reference firmware for the MSP430FR2355, ESP32, STM32F103C8, and RP2040, each emitting the same JSON-over-UART frame format so any of the four boards works with the same bridge and backend.
- Built the Python MQTT bridge and a hardware simulator, plus a capture-and-replay tool for running demos and regression tests without physical hardware.
- Added schema-validated frame parsing and a cross-platform test suite checking that all four firmware builds produce frames the backend accepts.
- Set up a GitHub Actions workflow that spins up Postgres and Mosquitto containers and runs an end-to-end simulated hardware test on every push.

Open Source

Open Source Contributions

2026

C++, Python, D-Bus, Linux, Gerrit code review

- OpenBMC phosphor-user-manager: diagnosed and fixed two bugs in a BMC user management daemon, an unhandled exception that could abort account setup for every user over one bad system record, and a C++ virtual dispatch issue causing unit tests to silently run against the real system instead of test mocks; submitted for review with full before and after test verification.
- Zephyr RTOS: found and fixed a devicetree macro bug in the MIPI DBI display driver that produced an incorrect chip select GPIO check for out of range devices; submitted with a compile time regression test.
- PlatformIO: diagnosed and fixed a crash in the serial device monitor triggered by non-interactive input; merged upstream with an added regression test.

Experience

City of Winter Haven Technology Services | *Smart City Student Volunteer* | Winter Haven, FL

April 2026 - Present

- Installed server hardware and performed rack maintenance for city technology infrastructure.
- Provisioned and configured YubiKey security keys for staff authentication.
- Resolved IT support tickets and assisted with early-stage AI hosting infrastructure.

Subs N' Such | *Manager, Shift Leader, and Front-End Web Support* | Lutz, FL

April 2023 - March 2026

- Built and maintained a Square-based website for online ordering.
- Managed shift operations and enforced food safety procedures.

Skills

Languages: C, C++, Python, Verilog/VHDL, RISC-V Assembly, MATLAB, Bash

Hardware: MSP430FR2355, Raspberry Pi, Arduino, FPGA (Xilinx), L298N, HM-10 BLE

Protocols: UART, I2C, SPI, PWM, MQTT, GPIO, Bluetooth LE

Tools: Git, Docker, Linux, Cursor, GDB, Xilinx ISE, SOLIDWORKS

Coursework: Embedded Systems, Computer Architecture, CMOS-VLSI Design, FPGA Design, Operating Systems, Computer System Design, Trustworthy Infrastructures

Certifications: Certified SOLIDWORKS Associate (CSWA), CSWA Additive Manufacturing

Involvement

Member, Society of Hispanic Professional Engineers (SHPE), Society of Asian Scientists and Engineers (SASE), and IEEE